

Fermi's golden rule

Find the transition rate $R_{1 \rightarrow 2}$ for the perturbing Hamiltonian

$$H_1(\mathbf{r}, t) = 2V(\mathbf{r}) \cos(\omega t + \phi)$$

§1

Let $\Psi(\mathbf{r}, t)$ be the following linear combination of the two eigenstates.

$$\Psi(\mathbf{r}, t) = c_1(t)\psi_1(\mathbf{r}) \exp(-i\omega_1 t) + c_2(t)\psi_2(\mathbf{r}) \exp(-i\omega_2 t)$$

We need to solve for $c_2(t)$ to find the transition rate $R_{1 \rightarrow 2}$.

Start with the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = H_0(\mathbf{r})\Psi(\mathbf{r}, t) + H_1(\mathbf{r}, t)\Psi(\mathbf{r}, t)$$

where

$$H_0(\mathbf{r})\Psi(\mathbf{r}, t) = E_1 c_1(t)\psi_1(\mathbf{r}) \exp(-i\omega_1 t) + E_2 c_2(t)\psi_2(\mathbf{r}) \exp(-i\omega_2 t)$$

Evaluate the time derivative.

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) &= i\hbar \frac{\partial c_1(t)}{\partial t} \psi_1(\mathbf{r}) \exp(-i\omega_1 t) + \boxed{\hbar\omega_1 c_1(t)\psi_1(\mathbf{r}) \exp(-i\omega_1 t)} \\ &\quad + i\hbar \frac{\partial c_2(t)}{\partial t} \psi_2(\mathbf{r}) \exp(-i\omega_2 t) + \boxed{\hbar\omega_2 c_2(t)\psi_2(\mathbf{r}) \exp(-i\omega_2 t)} \\ &= H_0(\mathbf{r})\Psi(\mathbf{r}, t) + H_1(\mathbf{r}, t)\Psi(\mathbf{r}, t) \end{aligned}$$

Noting that $\hbar\omega_1 = E_1$ and $\hbar\omega_2 = E_2$, the boxed terms cancel with $H_0(\mathbf{r})\Psi(\mathbf{r}, t)$ leaving

$$i\hbar \frac{\partial c_1(t)}{\partial t} \psi_1(\mathbf{r}) \exp(-i\omega_1 t) + i\hbar \frac{\partial c_2(t)}{\partial t} \psi_2(\mathbf{r}) \exp(-i\omega_2 t) = H_1(\mathbf{r}, t)\Psi(\mathbf{r}, t)$$

Substitute for $H_1(\mathbf{r})$ and $\Psi(\mathbf{r}, t)$.

$$\begin{aligned} i\hbar \frac{\partial c_1(t)}{\partial t} \psi_1(\mathbf{r}) \exp(-i\omega_1 t) + i\hbar \frac{\partial c_2(t)}{\partial t} \psi_2(\mathbf{r}) \exp(-i\omega_2 t) \\ = 2V(\mathbf{r}) \cos(\omega t + \phi) \left[c_1(t)\psi_1(\mathbf{r}) \exp(-i\omega_1 t) + c_2(t)\psi_2(\mathbf{r}) \exp(-i\omega_2 t) \right] \end{aligned}$$

Take the inner product of $\psi_2^*(\mathbf{r})$ with both sides to obtain

$$i\hbar \frac{\partial c_2(t)}{\partial t} \exp(-i\omega_2 t) = 2 \cos(\omega t + \phi) \left[c_1(t)M_{21} \exp(-i\omega_1 t) + c_2(t)M_{22} \exp(-i\omega_2 t) \right]$$

where M_{21} and M_{22} are the matrix elements

$$\begin{aligned} M_{21} &= \int \psi_2^*(\mathbf{r}) V(\mathbf{r}) \psi_1(\mathbf{r}) d\mathbf{r} \\ M_{22} &= \int \psi_2^*(\mathbf{r}) V(\mathbf{r}) \psi_2(\mathbf{r}) d\mathbf{r} \end{aligned}$$

Multiply both sides by $\exp(i\omega_2 t)$.

$$i\hbar \frac{\partial c_2(t)}{\partial t} = 2 \cos(\omega t + \phi) \left[c_1(t) M_{21} \exp[i(\omega_2 - \omega_1)t] + c_2(t) M_{22} \right]$$

Let the initial state be $\Psi(\mathbf{r}, 0) = \psi_1(\mathbf{r})$ hence the initial conditions are

$$c_1(0) = 1, \quad c_2(0) = 0$$

For time t near the origin use $c_1(t) \approx 1$ and $c_2(t) \approx 0$ to obtain

$$i\hbar \frac{\partial c_2(t)}{\partial t} = 2 \cos(\omega t + \phi) M_{21} \exp[i(\omega_2 - \omega_1)t]$$

Solve for $c_2(t)$ by integrating.

$$c_2(t) = \frac{2M_{21}}{i\hbar} \int_0^t \cos(\omega t' + \phi) \exp[i(\omega_2 - \omega_1)t'] dt'$$

In exponential form

$$\begin{aligned} c_2(t) = \frac{M_{21}}{i\hbar} \int_0^t \exp(i\omega t' + \phi) \exp[i(\omega_2 - \omega_1)t'] dt' \\ + \frac{M_{21}}{i\hbar} \int_0^t \exp(-i\omega t' - \phi) \exp[i(\omega_2 - \omega_1)t'] dt' \end{aligned}$$

The solution is

$$\begin{aligned} c_2(t) = -\frac{M_{21}}{\hbar} \frac{\exp[i(\omega_2 - \omega_1 + \omega)t] - 1}{\omega_2 - \omega_1 + \omega} \exp(i\phi) \\ - \frac{M_{21}}{\hbar} \frac{\exp[i(\omega_2 - \omega_1 - \omega)t] - 1}{\omega_2 - \omega_1 - \omega} \exp(-i\phi) \quad (1) \end{aligned}$$

For ω such that $\omega \approx \omega_2 - \omega_1$ the second term dominates so discard the first term.

$$c_2(t) = -\frac{M_{21}}{\hbar} \frac{\exp[i(\omega_2 - \omega_1 - \omega)t] - 1}{\omega_2 - \omega_1 - \omega} \exp(-i\phi)$$

By the identity

$$\exp(ix) - 1 = 2i \sin\left(\frac{x}{2}\right) \exp\left(\frac{ix}{2}\right)$$

we have

$$c_2(t) = -\frac{2iM_{21}}{\hbar} \frac{\sin\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right] \exp\left[\frac{1}{2}i(\omega_2 - \omega_1 - \omega)t\right]}{\omega_2 - \omega_1 - \omega} \exp(-i\phi)$$

Multiply numerator and denominator by $t/2$.

$$c_2(t) = -\frac{itM_{21}}{\hbar} \frac{\sin\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right] \exp\left[\frac{1}{2}i(\omega_2 - \omega_1 - \omega)t\right]}{\frac{1}{2}(\omega_2 - \omega_1 - \omega)t} \exp(-i\phi)$$

Substitute $\text{sinc}(x)$ for $\sin(x)/x$.

$$c_2(t) = -\frac{itM_{21}}{\hbar} \text{sinc}\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right] \exp\left[\frac{1}{2}i(\omega_2 - \omega_1 - \omega)t\right] \exp(-i\phi) \quad (2)$$

§2

Transition density P is squared magnitude of $c_2(t)$.

$$P = |c_2(t)|^2 = \frac{t^2}{\hbar^2} |M_{21}|^2 \text{sinc}^2 \left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t \right] \quad (3)$$

Total density P_{tot} is

$$P_{tot} = \frac{t^2}{\hbar^2} |M_{21}|^2 \int_{E-\epsilon}^{E+\epsilon} \text{sinc}^2 \left(\frac{E'/\hbar - \omega}{2} t \right) \rho(E') dE'$$

where $E \pm \epsilon = \hbar(\omega_2 - \omega_1) \pm \epsilon$ and ρ is density of states.

Use the approximation $\rho(E') \approx \rho(\hbar\omega)$ to obtain

$$P_{tot} = \frac{t^2}{\hbar^2} |M_{21}|^2 \rho(\hbar\omega) \int_{E-\epsilon}^{E+\epsilon} \text{sinc}^2 \left(\frac{E'/\hbar - \omega}{2} t \right) dE'$$

Change variable to y where

$$y = \frac{E'/\hbar - \omega}{2} t$$

It follows that

$$E' = \frac{2\hbar y}{t} + \hbar\omega$$

and

$$dE' = \frac{2\hbar}{t} dy$$

The integration limits transform as

$$E \pm \epsilon \rightarrow \frac{(E \pm \epsilon)/\hbar - \omega}{2} t = \frac{Et}{2\hbar} - \frac{\omega t}{2} \pm \frac{\epsilon t}{2\hbar} = \pm \frac{\epsilon t}{2\hbar}$$

Hence

$$P_{tot} = \frac{2t}{\hbar} |M_{21}|^2 \rho(\hbar\omega) \int_{-\epsilon t/2\hbar}^{\epsilon t/2\hbar} \text{sinc}^2 y dy$$

Use the approximation

$$\int_{-\epsilon t/2\hbar}^{\epsilon t/2\hbar} \text{sinc}^2 y dy \approx \int_{-\infty}^{\infty} \text{sinc}^2 y dy = \pi$$

to obtain

$$P_{tot} = \frac{2\pi t}{\hbar} |M_{21}|^2 \rho(\hbar\omega)$$

The transition rate is the derivative of P_{tot} .

$$R_{1 \rightarrow 2} = \frac{d}{dt} P_{tot} = \frac{2\pi}{\hbar} |M_{21}|^2 \rho(\hbar\omega) \quad (4)$$

§3

We started with

$$H_1(\mathbf{r}, t) = 2V(\mathbf{r}) \cos(\omega t + \phi)$$

and obtained Fermi's golden rule (4). Consequently for $H_1(\mathbf{r}, t)$ such that

$$H_1(\mathbf{r}, t) = V(\mathbf{r}) \cos(\omega t + \phi)$$

we would have

$$R_{1 \rightarrow 2} = \frac{2\pi}{\hbar} \left| \frac{M_{21}}{2} \right|^2 \rho(\hbar\omega)$$